9[Page Replacement Policy]

A system has RAM which can store four page simultaneously to satisfy page requests. Write a program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page reference entered by the user i.e. 1,2,3,4,1,2,5,1,2,3,4,5 (No. of Frame 3).

CODE:

```
#include <stdio.h>

int main()
{
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frames[3];
    int i, j, k, count = 12;
    int found, flag;
    int page_fault = 0, page_hit = 0;
    int pos;

    // initialize frames
    for(i = 0; i < 3; i++)
        frames[i] = -1;

    for(i = 0; i < count; i++)
    {
        found = 0;

        // check page hit
        for(j = 0; j < 3; j++)
        {
            if(frames[j] == pages[i])
            {
                found = 1;
                page_hit++;
                break;
            }
        }

        if(found == 0)
        {
            if(i < 3)
            {
                frames[i] = pages[i];
            }
            else
            {
                int farthest = -1, replace = -1;
```

```
                for(j = 0; j < 3; j++)
                {
                    flag = 0;

                    for(k = i + 1; k < count; k++)
                    {
                        if(frames[j] == pages[k])
                        {
                            if(k > farthest)
                            {
                                farthest = k;
                                replace = j;
                            }
                            flag = 1;
                            break;
                        }
                    }

                    if(flag == 0)
                    {
                        replace = j;
                        break;
                    }
                }

                frames[replace] = pages[i];
            }

            page_fault++;
        }

        printf("Frames: ");
        for(j = 0; j < 3; j++)
            printf("%d ", frames[j]);
        printf("\n");
    }

    printf("\nTotal Page Faults = %d", page_fault);
    printf("\nTotal Page Hits = %d", page_hit);

    printf("\nPage Fault Percentage = %.2f", (page_fault/(float)count)*100);
    printf("\nPage Hit Percentage = %.2f\n", (page_hit/(float)count)*100);

    return 0;
}
```

OUTPUT:

```
MAYUR LAP MSYS ~
$ nano optimal.c

MAYUR LAP MSYS ~
$ gcc optimal.c -o optimal

MAYUR LAP MSYS ~
$ ./optimal
Frames: 1 -1 -1
Frames: 1 2 -1
Frames: 1 2 3
Frames: 1 2 4
Frames: 1 2 4
Frames: 1 2 5
Frames: 1 2 5
Frames: 1 2 5
Frames: 3 2 5
Frames: 4 2 5
Frames: 4 2 5

Total Page Faults = 7
Total Page Hits = 5
Page Fault Percentage = 58.33
Page Hit Percentage = 41.67

MAYUR LAP MSYS ~
```